



SIMATS ENGINEERING

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Course Code: MLA0408

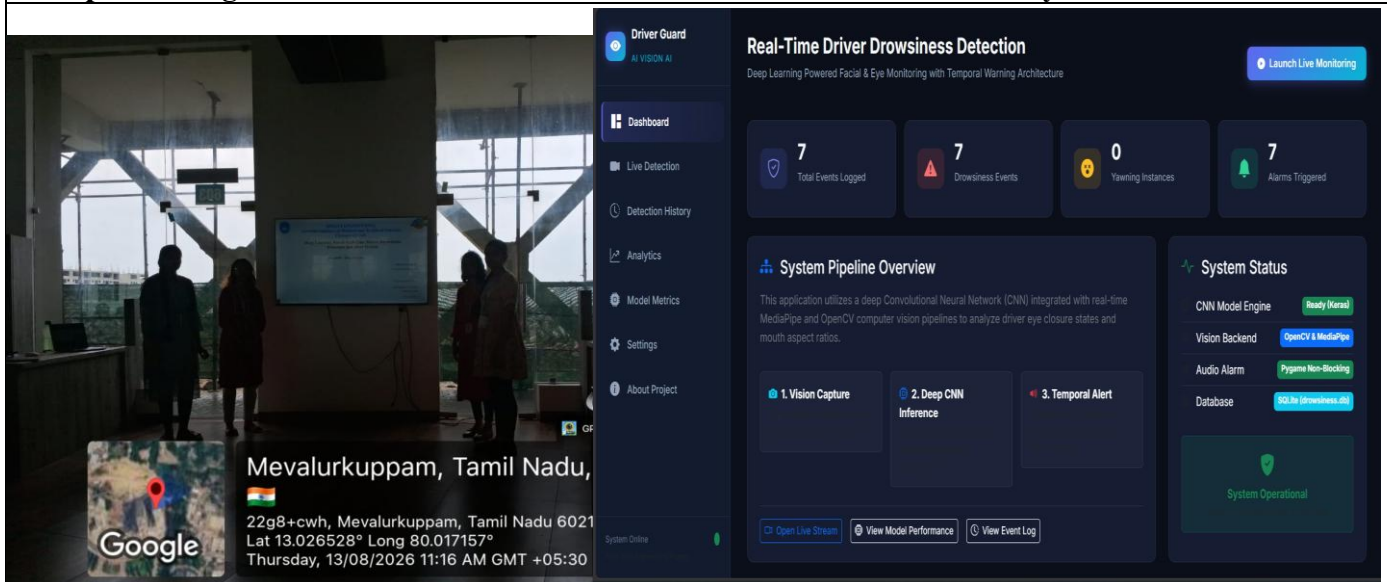
Slot: C

Course Name: Deep Learning

Course Faculty: Dr.D Sudhagar

Project Title:

Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System.



Project Description:

The **Deep Learning Based Real-Time Driver Drowsiness Detection and Alert System** is designed to detect whether a driver is alert or drowsy using a camera and deep learning. The system continuously captures the driver's face and analyzes features such as eye closure, blinking, and yawning. When drowsiness is detected, the system provides an immediate alert to warn the driver and improve road safety..

Module: Data Collection and Preprocessing

This module collects driver images and video frames containing **alert and drowsy conditions**, including open eyes, closed eyes, blinking, and yawning. The collected data is organized into suitable categories for training and testing the deep learning model.

The collected images are then preprocessed by **removing unwanted or poor-quality images, resizing the images, normalizing pixel values, and preparing the data in a suitable format**. This ensures that clean and consistent input is provided to the deep learning model for accurate drowsiness detection.

Student Signature

Guide Signature